F FWallSim · Firewall Course

CLASS 10

Site-to-site IPsec VPN

IPsec phase 1/2, routes, VPN policies

Advanced | ~30 minutes | FWallSim v1.0.0 (FortiOS 7.4 style)

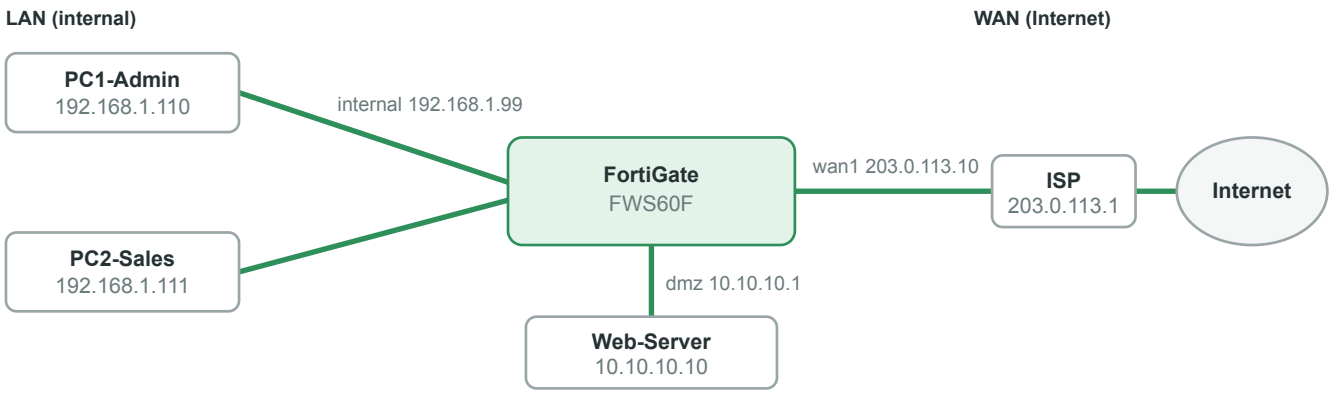
YOU WILL LEARN

- IPsec phase 1/2
- routes
- VPN policies

BEFORE YOU START

Finish Class 9 first.

The lab network



The goal

THE IDEA

Connect HQ to the branch office (Branch-FGT 198.51.100.200, LAN 10.20.0.0/24). The branch settings are fixed and shown in Lab > VPN: match them exactly.

How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **Site-to-site IPsec VPN** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

Tasks

1. **A tunnel to the branch is Up**

   VPN > IPsec Wizard (or a custom tunnel). Check VPN > IPsec Tunnels: the Reason column explains failures.

2. **PC1 can ping Branch-PC (10.20.0.10)**

   Needs a route to 10.20.0.0/24 via the tunnel and a policy internal → tunnel (NAT off).

3. **Branch-PC can RDP to PC1**

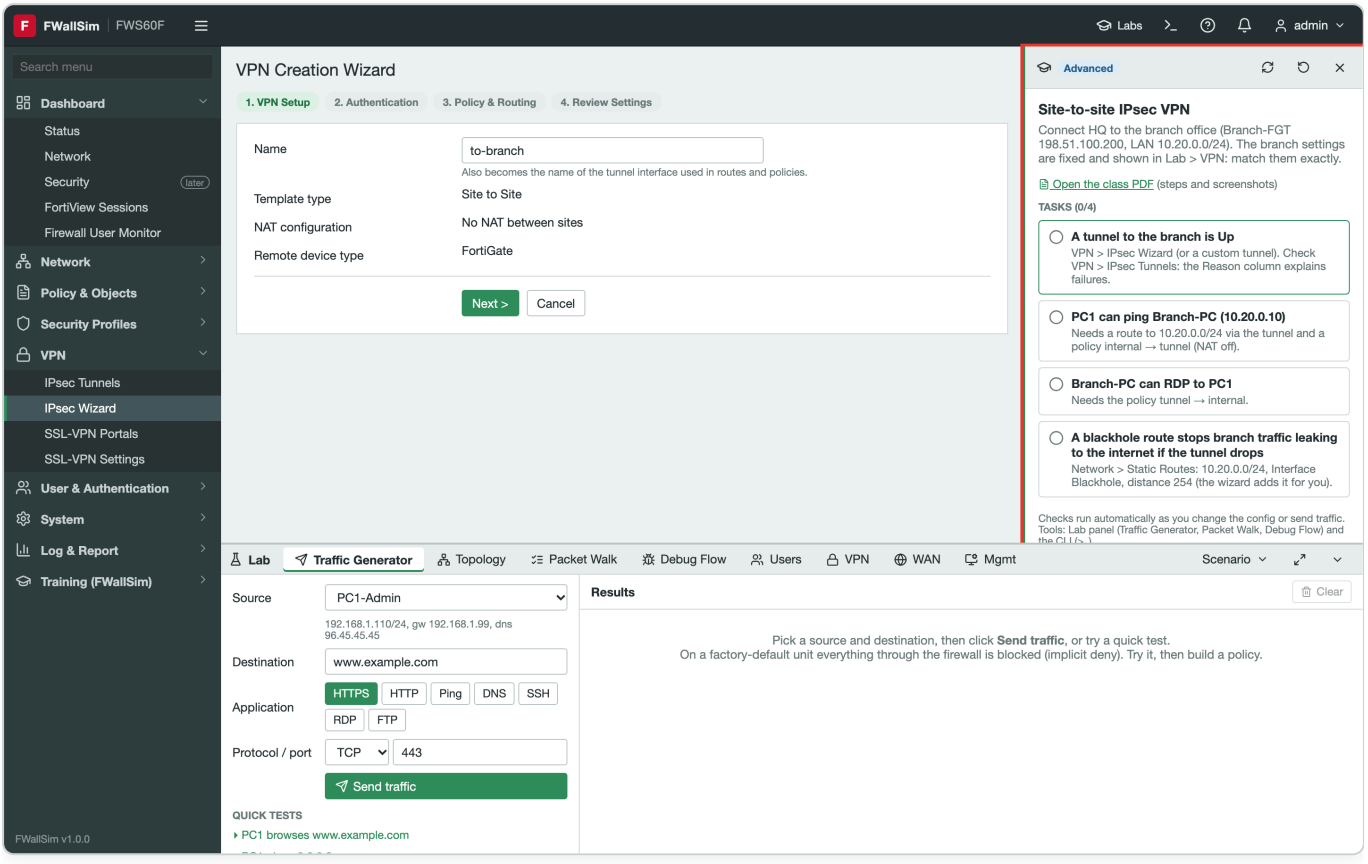
   Needs the policy tunnel → internal.

4. **A blackhole route stops branch traffic leaking to the internet if the tunnel drops**

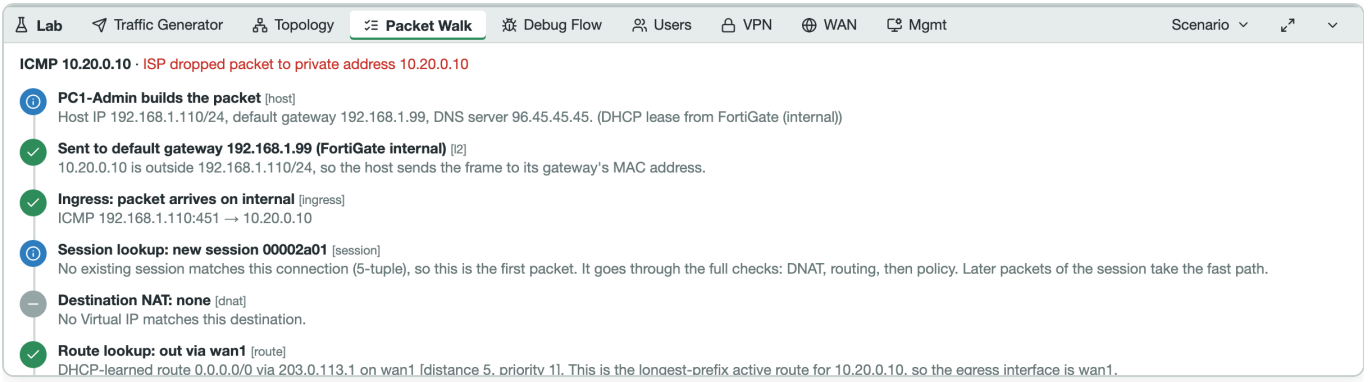
   Network > Static Routes: 10.20.0.0/24, Interface Blackhole, distance 254 (the wizard adds it for you).

Screenshots

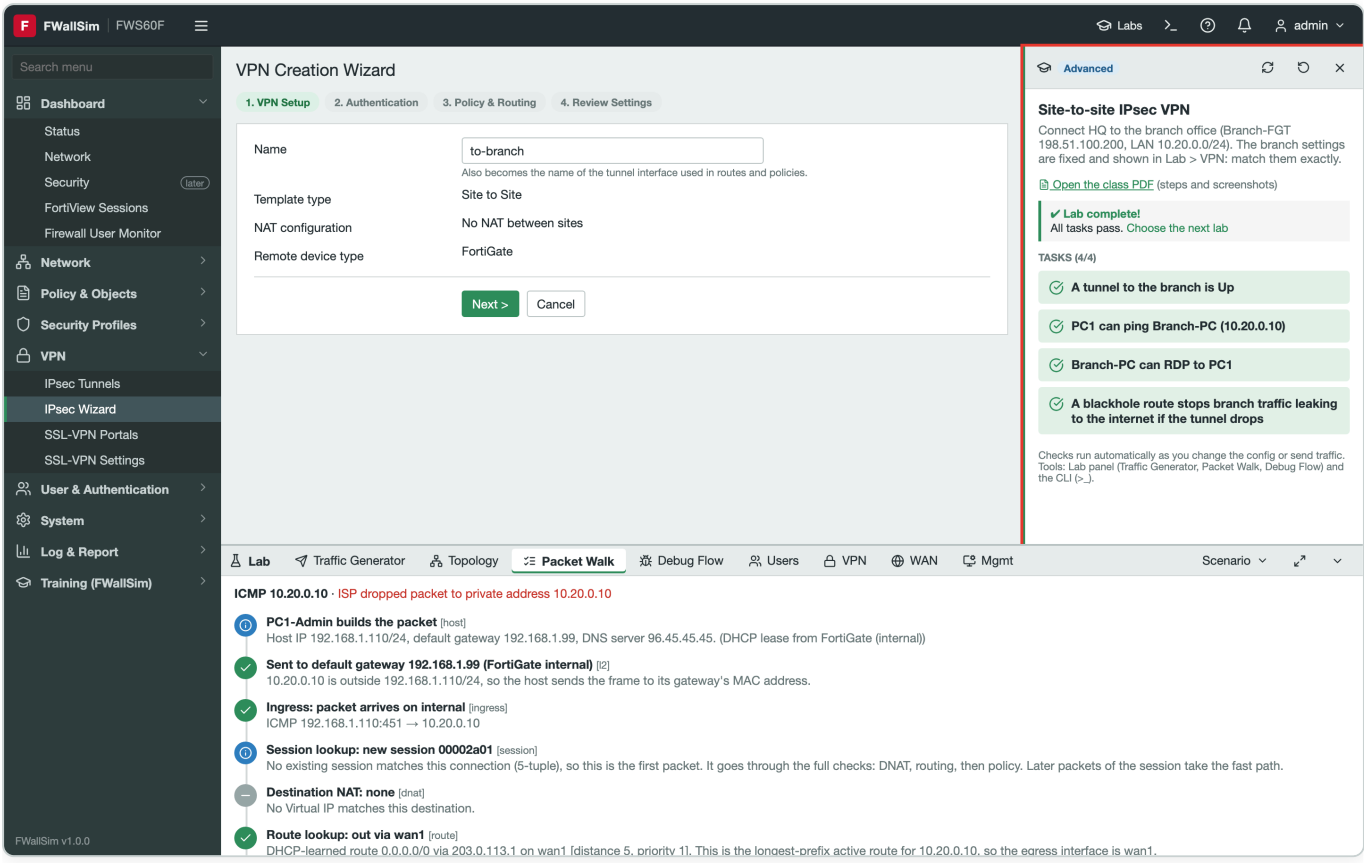
Numbered red markers show where to click or what to look at.



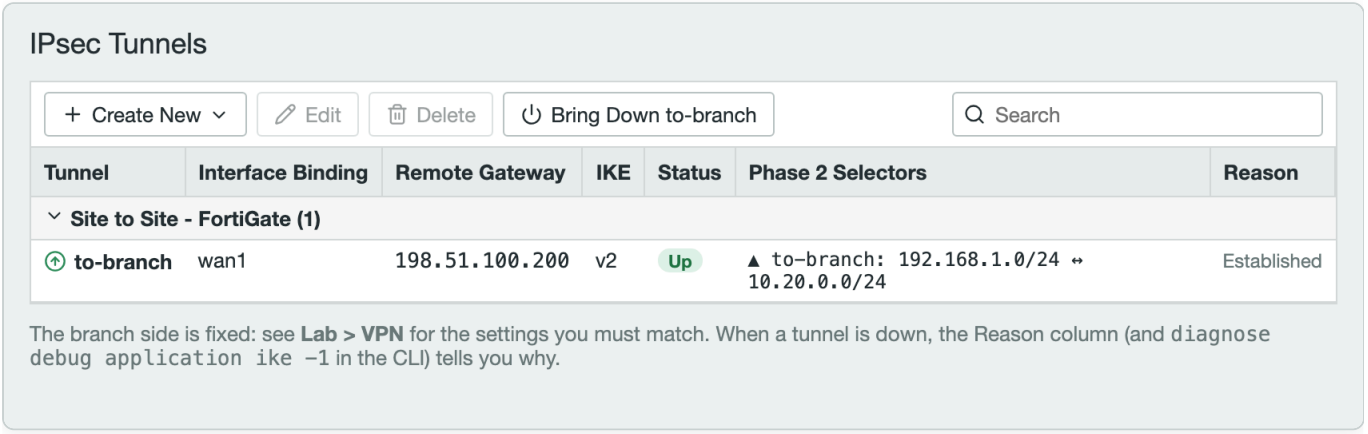
Lab Started



Packet Walk Before



Lab Solved



IPsec Tunnels

Static Routes

+ Create New | Edit | Clone | Delete | Search

Destination	Gateway IP	Interface	Distance	Priority	Status	Comme
10.20.0.0/24	0.0.0.0	to-branch	10	1	Active	
10.20.0.0/24		Blackhole	254	1	Inactive: higher distance than another route to the same prefix	

The default route learned by DHCP on wan1 does not appear here. See it under Dashboard > Network > Routing. Lower distance wins for the same prefix; for equal distance, lower priority wins.

Static Routes

Firewall Policy

+ Create New | Edit | Delete | Policy Lookup | Interface Pair View | By Sequence

Search

Name	ID	From	To	Source	Destination	Schedule	Service
internal → wan1 (1)							
LAN-to-Internet	1	internal	wan1	LAN-Subnet	all	always	ALL
wan1 → dmz (1)							
Web-Server-Inbound	2	wan1	dmz	all	VIP-WebServer-HTTP	always	HTTP
dmz → wan1 (1)							
DMZ-to-Internet	3	dmz	wan1	WebServer-DMZ	all	always	DNS HTTP HTTPS
internal → to-branch (1)							
hq-br	4	internal	to-branch	HQ	BR	always	ALL
to-branch → internal (1)							
br-hq	5	to-branch	internal	BR	HQ	always	ALL
Implicit (1)							
Implicit Deny	0	any	any	all	all	always	ALL

Policies are evaluated from top to bottom and the **first match wins**. Drag rows to reorder them. Right-click for more actions (insert, clone, enable/disable, logs). Right-click the Implicit Deny row to log violation traffic.

Firewall Policy

Lab | Traffic Generator | Topology | **Packet Walk** | Debug Flow | Users | VPN | WAN | Mgmt | Scenario

ICMP 10.20.0.10 · PC1-Admin → Branch-PC ICMP echo: delivered and answered, allowed by policy 4

PC1-Admin builds the packet [host]
Host IP 192.168.1.110/24, default gateway 192.168.1.99, DNS server 96.45.45.45. (DHCP lease from FortiGate (internal))

Sent to default gateway 192.168.1.99 (FortiGate internal) [l2]
10.20.0.10 is outside 192.168.1.110/24, so the host sends the frame to its gateway's MAC address.

Ingress: packet arrives on internal [ingress]
ICMP 192.168.1.110:914 → 10.20.0.10

Session lookup: new session 00002a02 [session]
No existing session matches this connection (5-tuple), so this is the first packet. It goes through the full checks: DNAT, routing, then policy. Later packets of the session take the fast path.

Destination NAT: none [dnat]
No Virtual IP matches this destination.

Route lookup: out via to-branch [route]
Static route 10.20.0.0/24 via direct on to-branch [distance 10, priority 1]. This is the longest-prefix active route for 10.20.0.10, so the egress interface is to-branch.

Packet Walk After

Next: Class 11: SSL-VPN for remote workers